

Framing the Unseen: Gesture, Network, and Intent in the Art of the Screenshot

Marty Miller

Centre for Applied Computing & Interactive Media: School of Creative Media
City University of Hong Kong
Kowloon, Hong Kong
martinmiller18@gmail.com

Abstract

This paper investigates screenshots as intentional or automated acts of device and network interaction. Their significance lies as much in how they are interpreted as in how they are produced. Pictured examples are analyzed for gestural, software, and networked influences that shape what becomes visible. Some highlight interpretive reflections on interactive processes, while others raise doubts about authorship when time-automated software produces similar results. To probe this tension, screenshots were introduced to local photographers, prompting reflections on normally unseen networked or embodied interactions within their practice. These findings suggest that screenshots, though often overlooked, can serve as probes for authorial intent, reframing how everyday images function as archives of both physical and networked interaction.

CCS Concepts

• Human-centered computing → Interaction design.

Keywords

screenshots, visual interaction, photographic intent, user agency, imaging software, network aesthetics

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1 Introduction

The screenshot owes its existence to the development of networked imaging technology and to the associated increase in pictures produced. Smartphone use in particular has contributed to this imaging technique's ubiquity among average device users. Yet the screenshot's extended signifying potential remains overlooked, with only a handful of critical works treating it or its utility as a distinctive cultural practice [Nešović 2022; Steyerl 2009]. Differing from traditional photography, the screenshot has a unique signifying potential. When viewing a set of these pictures, the embodied and extended

network interaction of the device user is directly inferred. In this paper, a series of these examples suggests how to frame [Goffman 1974] the visual content for deeper analysis [Bock 2020]. These are followed by photographers' initial reflections on this medium's utility to probe how photographic authorship is recognized.

To begin, a visual content analysis of the screenshot will suggest how on-screen content can capture unexpected moments, such as when text overlaps with the image, producing new means to access the screenshot creator's intent to preserve information [van Lier 2007]. As an aid, a discursive design method is employed to stimulate discussion about this intent by juxtaposing manually taken examples with those generated by time-automation software [Tharp and Tharp 2019]. Accordingly, understanding how these screenshots are created involves differentiating between human and algorithmic ways of making meaning. The contrast between these readings highlights reflective questions needed to recognize a screenshot's human author. Examples embodying this tension are then utilized to probe the reactions of practicing photographers and visual designers. Their discussion suggests noting on-screen content as a form of archiving with distinct signifying potential [Rinehart and Ippolito 2022]. Combining picture-taking with on-screen content curation [Bock 2020] implicates not just the networks recontextualizing content, but also the embodied gestures and discernible intentions which give the screenshot meaning [Zylinska 2017].

2 Framing the Gesture: The Unseen Bodily Act

Although often overlooked in the discourse of photographic art and visual design, contemporary media theorists have discussed the screenshot's expansive network and gesturally reflective potential. Recent writing suggests the medium affords secondary framing of common photos in daily life [Miller 2024; Švelch 2021]. Also discussed is what helps screenshots signify past connections between body and device or network interaction [Feiz et al. 2022]. Discerning such potential rests on an important distinction.

First, a screenshot can be considered different from a screen capture because it depicts only what is on the screen [Švelch 2021]. In other words, there is no difference between the picture and the screen's boundaries. What is visible in the screenshot resembles what is visible through the viewfinder. Compared to a screen capture, a screenshot would show a photo of a screen taken by a separate device. Therefore, a screen capture would likely contain some visible context as pictured in the area around the screen [Miller 2023]. Content in a screen capture is more directly world-facing, whereas a screenshot's content is facing directly inward to the software that enables its display.

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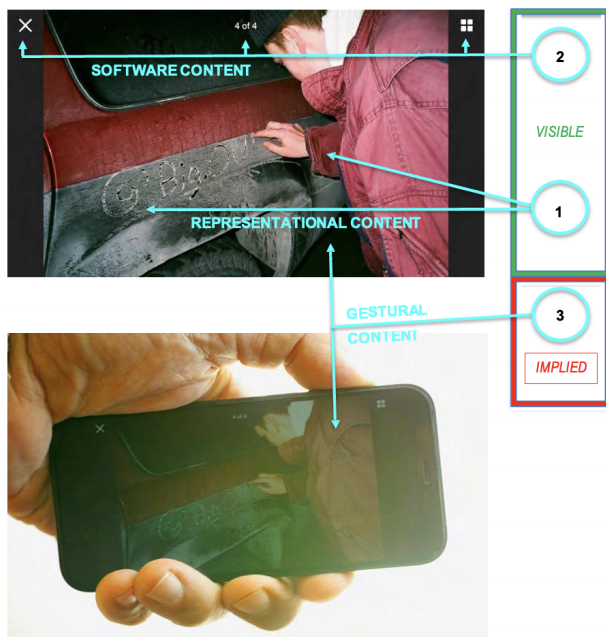


Figure 1: Two photos with visible representational and software content, and gestural implications.

Viewing a screenshot implicates an embodied user's direct decision [Ihde 2009] to hold the device and, like in traditional photography [Cartier-Bresson 1952], choose the moment to capture the content on the screen [Frosh 2018]. While the embodied interaction with the device is not visible, it is implicated by the mere existence of the screenshot, as pictured in Figure 1 [Flusser 2014]. Similarly, software within the imaging device allows screenshots to retain information that the device's user deems valuable. As designers of such software are likely also its users, the screenshot shows how common and accessible this act of image-making has become. For example, simultaneously pressing two buttons (as in the bottom picture within Figure 1) suggests ease of production.

The device's imaging software also makes it easy to catalog photographs as information to remember and like or dislike (as in Figure 2). Together, the design of the screenshot as a tool for information archiving and retrieval reveals how accessible the connection between information retention and image making has become [Frosh 2018].

Both the everyday and unusual are signified when performing a visual content analysis of the screenshot. Figure 1 shows three different ways to read content. Together, they support analysis of why and how the screenshot was taken. Points of interest may infer what is usually considered valuable content of a photo, but also extend beyond this [Flusser 1984]. Included are: (1) the representational, (2) the internal software, and (3) the external embodied gesture.

Figure 2 shows what users of the imaging device can notice about the motivation of the screenshot producer [Mottelson 2023]. Unless fully automated, it is often taken for granted that an embodied

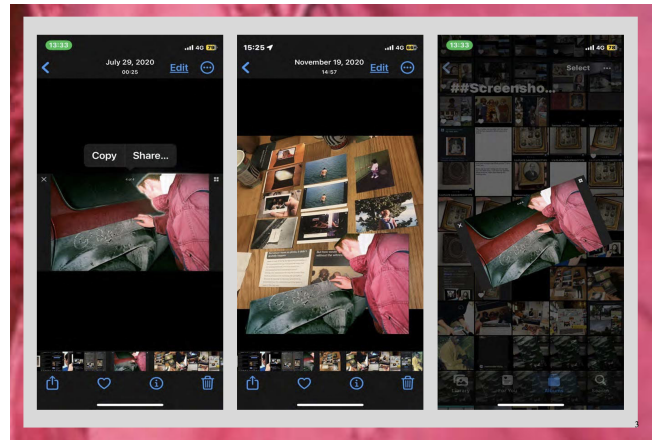


Figure 2: Three screenshots taken in succession. Each highlights the screen being touched, dragged, or rotated.

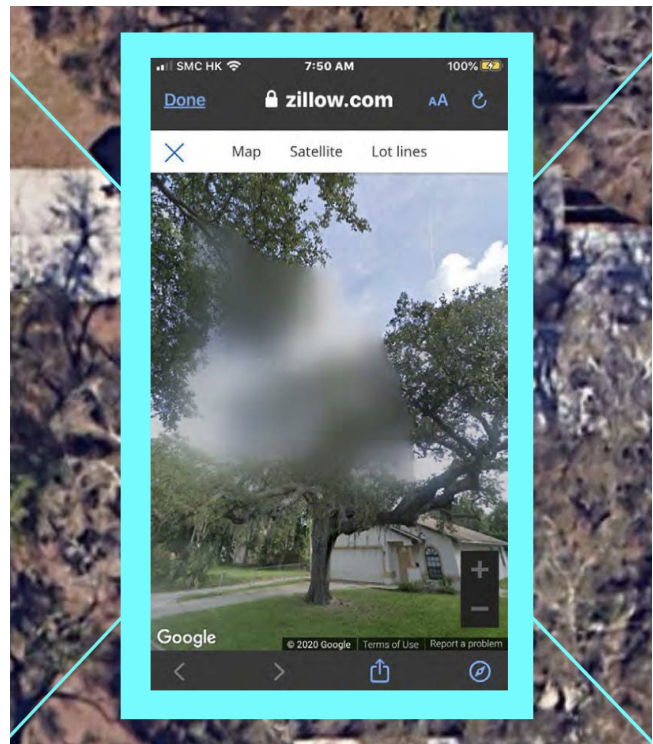


Figure 3: Screenshot of map software showing the center area concealed without a clear indication of the cause.

gesture is needed to create these types of pictures [Flusser 2014]. However, the contribution of screenshots lies in observing types of interrelated content that *can* be made visible [Verbeek 2017]. As pictured through the photographer's bodily contortion, prior interactions with photos become visibly contingent on the depicted software in the newly formed screenshot.

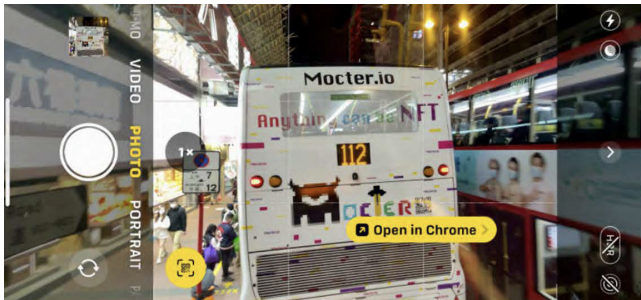


Figure 4: Screenshot of network interaction through an augmented live camera view with a software application and an embedded link overlaid on the scene.

3 Framing the Network: Visualizing Unseen Data

The following images extend the visual content analysis of the screenshot beyond the embodied interaction of the gesture, toward the network of influences connecting through the device itself [Verbeek 2017]. These include reproductions of existing locations, such as those found in mapping software, as well as real-world content augmented by software that is made visible on the screen. These indicate a synchronized interaction between network content, the device, and the device user's preferences for memorable content [Gamboa et al. 2023]. Accordingly, reflections range from noticing the speed of content transfer to discovering unexpected meanings between seemingly unrelated content.

3.1 Concealing

Once recognized, a screenshot's visual content signifies what is concealed through the device's connection to a more extensive network, correlating and displaying data for on-screen use. Yet by outputting a still image, the screenshot suggests limitations on what can be seen about the action requested of the software by the device user. The value of unseen content becomes highlighted through the screenshot's affording both visible and invisible layers (as noted in Figure 1). For example, the screenshot, by design, usually conceals the gesture that created it. Yet, content within the screen's surface area can represent concealment more directly.

The screenshot in Figure 3 represents concealment within the frame itself. Taken while browsing a popular mapping application, attention is quickly drawn to the picture's large central area. Framed by tree branches and a house primarily in focus, this blurriness is deemed critical due to the opacity of its content. This suggests potential for additional information retrieval through the software's connection to a data-production network beyond the frame. A similar phenomenon is noticed in Figure 4. This screenshot faces outward into the physical world of a bus in transit, with portions of the screen concealed by software links. However, the site's additional information is not shown. These examples suggest that screenshots conceal information by augmenting the screen surface through the transfer of pictures across devices or networks.

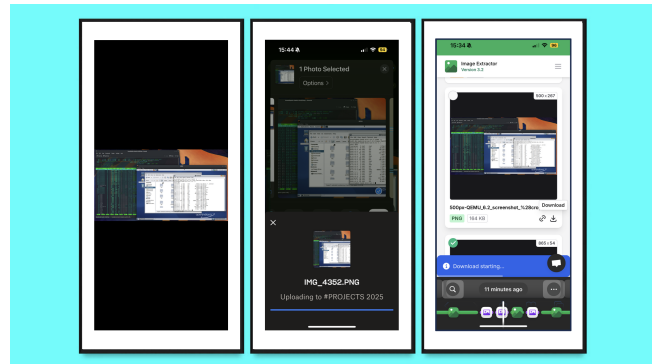


Figure 5: A comparative set of intentionally produced and time-automated screenshots both shows similar content, such as an image upload process.

3.2 Revealing

The screenshot can suggest meaning recognized through data consumption, especially when shown in progression. As in a minimally designed portfolio-as-probe, Figure 5 juxtaposes intentionally with those automatically produced through time-automated screenshot software. Visualized from the left, the image seems to show the downloaded picture in the browser viewer. However, reflection on image-sharing itself is signified by the picture in the cloud as portrayed in the center. Moving to the right, this relationship between automatic and human authorship is complicated. The rightward picture reveals the automated software as noticeable, thereby revealing latent assumptions about recognizing human intent. Their visual similarity reveals the difficulty in discerning one type from the other. However, if viewed as a set intentionally produced for specific meaning, visual content analysis reveals interaction in two directions [Elsden et al. 2017]. One shows a relationship of a single phone to the data network. The other reveals the device user's gestural relation to their phone.

In either case, screenshots can reveal the relation of the imaging device to the network. They do this through juxtaposing software and representational content in the original photograph. Figure 6 shows this potential for unplanned meaning to occur between the network completing a download and the construction project nearing completion. This link is made clear through the connection of the arrows added to the screenshot during post-production [Elsden et al. 2017]. Figure 7 shows a similar relation. Yet here, the message, "Wow, looking forward," reinforces the meaning of the upload process, a project's partial completion. Accordingly, these are likely to have been intentionally produced.

Figures 6 and 7 reveal a likely serendipitous experience for the device user. In Figure 6, the incident was triggered by noticing a correlation between visual and textual content, contingent on the network's speed. Figure 7 reveals a device user's appreciation for unexpected meaning-making from content on the screen. Here, meaning is contingent upon the network speed and the likelihood of receiving a text message. Not only did the message occur at the moment of upload, but it also, by chance, the message's content,

“looking forward,” seemed to reference the impending completion of the upload process.

Together, figures 3-7 show external network interaction. Yet, the choice of a moment to capture in Figures 6 and 7 reveals a screenshot-maker’s noticing of meaning. The screenshot conveys this human intention to preserve meaning for a viewer who can, ideally, see how the screenshot-taker perceived it. Reflecting a persistent trend in contemporary imaging, such an observation is of the second order: a screenshot viewer observing how a screenshot maker viewed on-screen content. This analytical stance inherently suggests a comparison of meanings [Luhmann 2000]. Meanwhile, comparing multiple similar experiences can occur interpersonally, such as in a discussion setting.

4 Framing Intent: A Dialogue on Photographic Practice

As shown in the previous figures, screenshots can display physical interaction with images through the confines of their screen-as-frame. However, when physically printed, they become helpful for larger group reflection. The previously displayed screenshots, and twenty more, were collected and printed for introduction to a group of Hong Kong photographers interested in visualizing interaction. The set was envisioned as a cultural probe [Gaver et al. 2004] to gauge how practitioners relate uncertain content readings to their practice [Verbeek 2017]. Screenshots were deliberately printed to resemble photographs, as participants were already familiar with this scale in the context of the photographic medium. To promote physical interaction, they were displayed on a large table as photographs used during an ideation phase for portfolio creation. The screenshots could also be collected in small groups (see Figure 8, left side) to elicit discussion [Blevis 2016; Gaver et al. 2004; Sengers et al. 2005].

In total, six participants, comprising four men and two women, attended a discussion session at a community arts space in Hong Kong. They were practicing photographers or visual designers, recruited through a long-standing community photography organization. Connection and consent were established before the meeting. Introduced in the manner of a design probe, participants were allowed to interact with the images in a non-hierarchical manner. The samples were not presented in any fixed order, which encouraged exploratory remarks during the two-hour discussion. Data were collected through a combination of audiovisual documentation, firsthand observations, and follow-up questions.

The discussion proceeded through several stages. First, an introductory conversation established that participants wanted to gain a deeper understanding of their own practices. Several speculated about what a generative AI system might learn from their personal imaging style. They also reflected on the everyday datasets created through the routine use of their phone cameras. Once accessed (Figure 8, center), the printed screenshots elicited reflection on the network or bodily influences needed to make such images [Baumer et al. 2014; Frosh 2018].

Participant responses revealed counterfactual and interpretive insights. One note of Figure 6: “This image of the software bar saving the new building seems like it could have been taken unintentionally, because it is so normal, why would anyone want to save

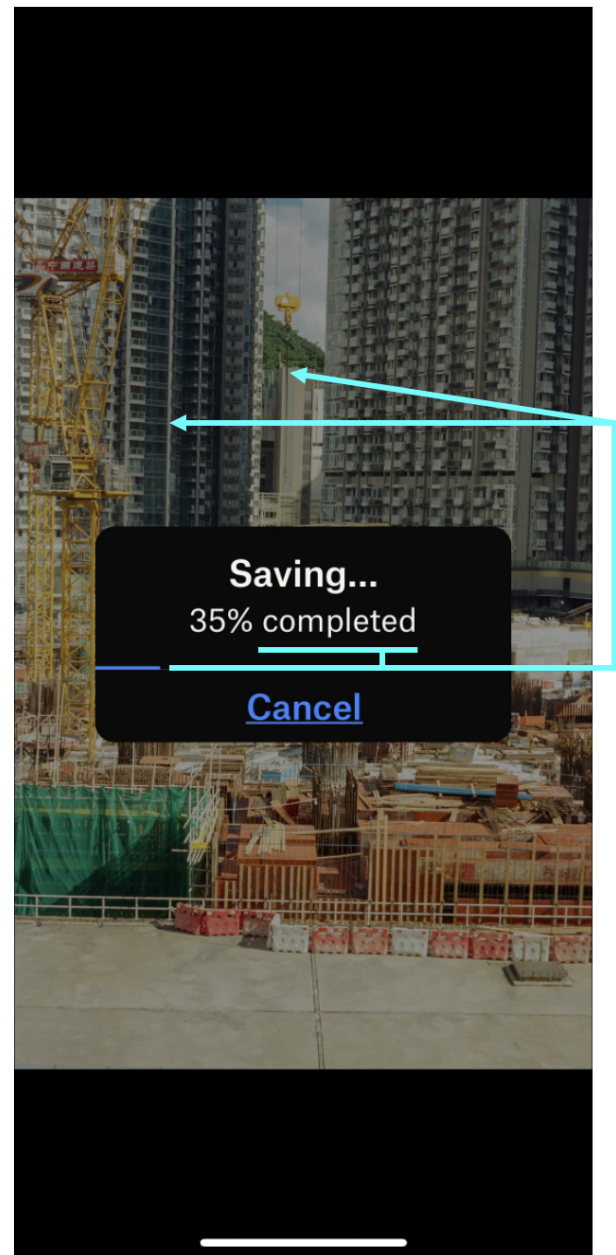


Figure 6: Screenshot showing the meaning correlation between a software icon (download process) and buildings.

that?” Another commented on Figure 7: “This ‘looking forward’ makes me wonder why I need to keep doing photography... maybe finding these types of beautiful moments by chance is most of the work.”

Others emphasized multimodal correlations between text and image components. One participant stated: “I want to make images that suggest a new building is as complete as the download or upload process, because I experience this every day but never

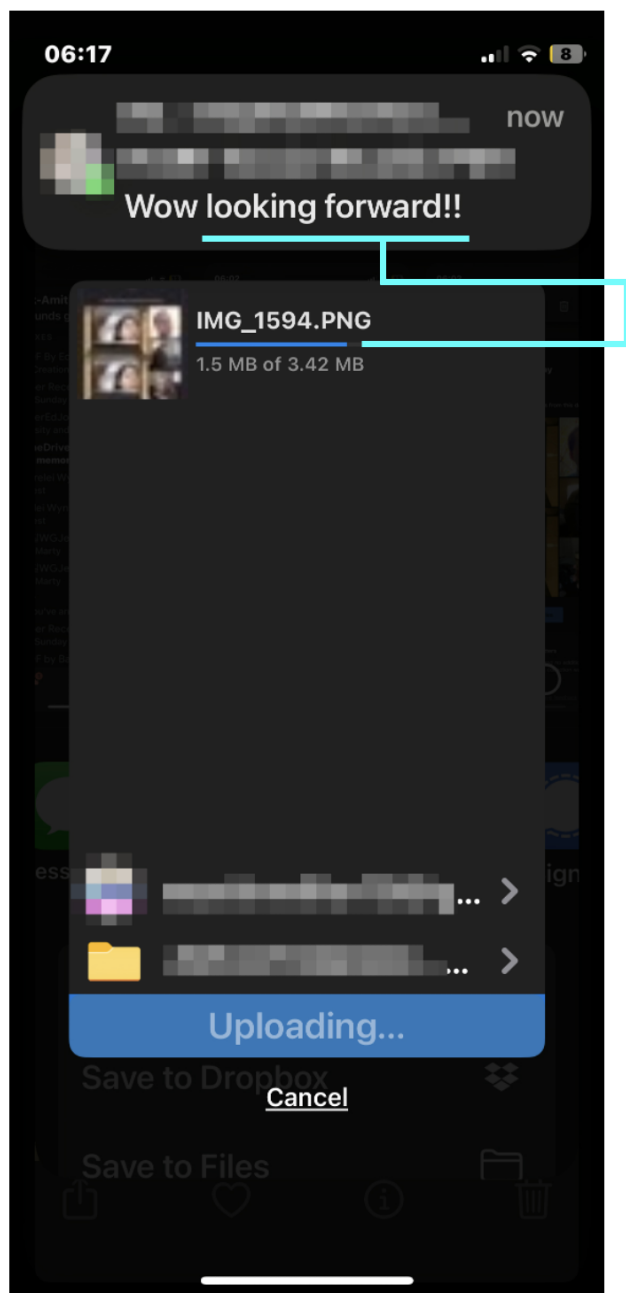


Figure 7: Screenshot showing a correlation in meaning between two software icons, the download process, and text message content.

think of the software icon as an expressive tool.” Another highlighted ambiguity in authorial intent: “It is best when my audience does not know if I knew what I was doing or not.” This remark was linked back to traditional photography with the observation that “this would happen when I choose the perfect, or ‘decisive’ moment to make a screenshot.” Collectively, these perspectives

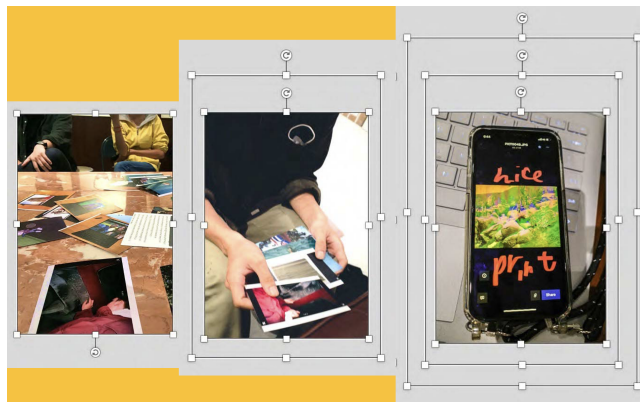


Figure 8: A sequential set of three pictures showing group discussion (left) followed by individual reflection (center) and finally a personal screenshot created to communicate how to determine if a screenshot was intentionally created.

suggest that screenshots can communicate subtle authorial intent while resonating with traditions of street photography.

In general, the comments of the participants revealed parallel authorial sensitivities between the photographic and the screen practices. Both rely on freezing time by pressing the right button and holding meaning-making objects within (or just outside) the frame. During the session, a quiet participant contributed their own screenshot (Figure 8, right). Featuring the words “nice print” bracketing a fluorescent picture of cloud storage, this example captured the event itself. It emphasized how intentionally produced screenshots can reveal bodily and software interactions in real-time.

These discussions highlight how screenshots elicit varying interpretations of shared daily life experiences [Gamboa et al. 2023]. Their subtle and complex signals of intention invite a creative reinterpretation of the unseen networks and embodied gestures that underpin digital imaging practices.

5 Conclusion

This paper has explored the screenshot as an act of framing, where on-screen content is concealed or revealed to suggest intentionality, even when its cause remains uncertain. Interpreting these discursive frames invites critical reflection on the interaction between the body, the imaging device, the device software, and the extended data network or physical world represented on the screen. In this way, the authorial intent of the human screenshot-maker is also signified through extended off-screen relations.

By presenting screenshots to practicing photographers, their value as prompts for creative discussion was demonstrated, revealing their potential to elicit reflection on unseen interactions in photographic practice. Further research can assemble these into collections to promote visually and technologically literate questioning of what makes on-screen content available and valuable to both the creator and the viewer. Ultimately, the art of the screenshot lies in the frame chosen to highlight the complex entanglement of network, gesture, and authorial intent.

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